Name each force and its source

LEARNING TARGET · K2 I can name each force, its source and its receiver.

SECTION A KEY VOCABULARY

Weight W	Earth's downward pull: mg.
Normal N	Surface push, perpendicular to contact.
Tension T	Rope's pull along itself, away from the object.
Receiver	The object a force acts on; also name its source.

SECTION B KEY EQUATIONS + UNITS

$$W = mg$$

$$N + T - W = 0$$

g	**9.8 N/kg**
balance	vertical $a = 0$

SECTION C COLOR IT · ONE CRATE, THREE FORCES

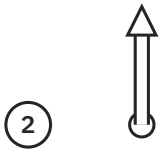
A rope helps a table.

rope on crate



T = 10 N

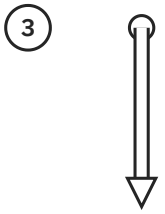
table on crate



N = 19.4 N

Earth on crate

W = 29.4 N



Three forces; one receiver. The crate stays at rest.

- ☐ **1 · ORANGE** Rope's tension T
- ☐ **2 · ORANGE** Table's normal N
- ☐ **3 · ORANGE** Earth's weight W

SECTION D WORKED EXAMPLE

A **3 kg** crate rests on a table. A vertical rope pulls up with **10 N**. **Find the table's normal force.** Use g = **9.8 N/kg**.

GIVEN	EQUATION	WORK	ANSWER
m = **3 kg** T = **10 N** up a_y = 0	$W = mg$. $N + T - W = 0$.	W = 3(9.8) = **29.4 N** N = 29.4 − 10	N = **19.4 N** The rope shares the support.

SECTION E THE 2026-XJ THREAD

DAY 10 OF 22 · NEW TOOL: FORCE SOURCES

A forecast of XJ's changing motion needs real interactions, not a made-up motion force. Name the attracting body for each gravitational force before adding the forces.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else